A SMART HELMET

FIELD OF THE INVENTION

[0001] The present invention relates to the field of road safety, and more specifically, the present invention relates to a smart helmet for collision detection, prevention, and vehicle ignition control.

BACKGROUND OF THE INVENTION

[0002] Every year, the number of road fatalities is increasing due to various factors such as distracted driving, speeding, drunk driving, and poor road conditions. Conventional helmets are primarily designed to provide passive protection, reducing head injuries in the event of an accident. However, they do not contribute to preventing accidents or ensuring that riders consistently wear helmets.

[0003] The helmet is not worn by many people due to a lack of awareness regarding its importance, discomfort during long rides, or the perception that it is unnecessary for short trips. Additionally, some riders may feel a false sense of security, believing that their riding skills alone can keep them safe. This mindset, combined with societal attitudes that downplay the risks associated with riding without proper headgear, further exacerbates the issue.

[0004] The helmet provides safety in the event of collisions or accidents, but it does not support the proactive measures needed to mitigate risks before they occur. Despite advancements such as GPS tracking and accident alert systems, existing smart helmets lack real-time preventive features and enforcement mechanisms. To address these challenges, there is a need for a smart helmet that can detect and prevent collisions, as well as control vehicle ignition.

OBJECTS OF THE INVENTION

[0005] Some objects of the present invention, which at least one embodiment herein satisfies, are as follows:

[0006] A Primary object of the present invention is to provide a smart helmet for collision detection, collision prevention, and vehicle ignition control.

[0007] Another object of the present invention is to detect approaching obstacle in real time.

[0008] Another object of the present invention is to prevent the ignition of vehicle until

[0009] Other objectives and advantages of the present invention will be more apparent from the following description, which is not intended to limit the scope of the present invention.

SUMMARY OF THE INVENTION

[0010] The following presents a simplified summary of the invention to provide a basic understanding of some aspects of the invention. This summary is not an extensive overview of the present invention. It is not intended to identify the key/critical elements of the invention or to delineate the scope of the invention. Its sole purpose is to present some concept of the invention in a simplified form as a prelude to a more detailed description of the invention presented later.

[0011] In an exemplary embodiment, the invention discloses a smart helmet which includes a helmet to be worn by a user for protection and operatively connected to an ignition interlock unit to detect if the helmet is worn by the user. A plurality of sensors operatively connected to the helmet to detect obstacles and impacts on the helmet. A microcontroller integrated in the helmet and configured to receive a sensor data, operation data from the ignition interlock unit, and location data, and perform analysis to identify approaching obstacles. An alert unit to provide a visual alert and an audio alert on a communication device when an obstacle is detected in a danger range. The microcontroller sends a signal via a communication network to the ignition interlock unit to lock the ignition of the vehicle if the helmet is not worn by the user.

[0012] In an implementation, the plurality of sensors comprises an accelerometer, a gyroscope, an ultrasonic sensor, and location sensor.

[0013] In another implementation, the smart helmet may include a rechargeable battery to supply power to the microcontroller, the plurality of sensors, and the communication device.

[0014] In another implementation, the smart helmet may include a retention unit with a buckle and a guard to detect if the helmet is worn by the user, wherein the retention unit is considered as closed when the buckle and the guard are connected with each other.

[0015] In yet another implementation, the smart helmet may include a messaging module to send an SMS alert to emergency contact of the user if the microcontroller detects vehicle collision based on the sensor data.

[0016] In yet another implementation, the microcontroller analyses the ultrasonic waves from the ultrasonic sensors of the plurality of sensors and calculate obstacle distance to alert the user if the obstacle distance is in the danger range.

[0017] In yet another implementation, the visual alert is shared on a user interface of the communication device and the audio alert is shared via a speaker on the communication device.

[0018] In yet another implementation, the microcontroller monitors the helmet usage of the user and display it on a user interface of the communication device.

[0019] These and other implementations, embodiments, processes, and features of the subject matter will become more fully apparent when the following detailed description is read with the accompanying experimental details. However, both the foregoing summary of the subject matter and the following detailed description of it represent one potential implementation or embodiment and are not restrictive of the present invention or other alternate implementations or embodiments of the subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] A clear understanding of the key features of the subject matter summarized above may be had by reference to the appended drawings, which illustrate the subject matter. However, it will be understood that such drawings depict preferred embodiments of the subject matter and, therefore, are not to be considered as limiting its scope with regard to other embodiments that the subject matter is capable of contemplating. It should also be noted that the accompanying figures are not necessarily drawn to scale. Accordingly:

[0021] **FIG. 1** illustrates a smart helmet, in accordance with an embodiment of the present invention; and

[0022] **FIG. 2** illustrates a flowchart for obstacle detection by the smart helmet, in accordance with an embodiment of the present invention.

[0023] The figures depict embodiments of the present subject matter for the purpose of illustration only. A person skilled in the art will easily recognize from the following description that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles of the invention described herein.

DETAILED DESCRIPTION OF THE INVENTION

[0024] The following is a detailed description of the implementation of the present invention depicted in the accompanying drawings. The implementation is in such detail as to clearly communicate the invention. However, the amount of detail offered is not intended to limit the implementation, but rather to cover all modifications, equivalents, and alternatives falling within the scope of the present invention. While aspects of the described invention can be implemented in any number of different configurations, the embodiments are described in the context of the following exemplary embodiment. Therefore, specific structural and functional details described herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the disclosed concepts in an appropriate structure or method. Furthermore, the terms and phrases used herein are not intended to be limiting, but rather to provide an understandable description of the present invention.

[0025] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of the ordinary skills in the art to which the invention pertains. The following description is provided with reference to the drawings, where like reference numerals are used to refer to like elements throughout. It should be noted that the accompanying figures are intended to present illustrations of exemplary embodiments of the present invention. In some instances, well-known structures and devices are shown in block diagram form in order to facilitate describing various techniques.

[0026] FIG. 1 illustrates a smart helmet (100) for collision detection, prevention, and ignition control. The smart helmet (100) is used by vehicle passengers, including two-wheeler riders and high-speed racing car drivers, who require a helmet for their safety. The vehicles that require this helmet include, but are not limited to, bikes and racing cars. The smart helmet (100) is equipped with Internet of Things (IoT) based sensors and a control unit to enforce helmet use, detect collisions, identify mishaps, and facilitate real-time emergency communication. It alerts the rider of approaching vehicles and provides real-time tracking, accident alerts, and Internet of Things connectivity through a communication device, such as a mobile application or a user interface. The smart helmet (100) promotes preventive safety, speeds up emergency response times, and encourages responsible riding behavior.

[0027] The smart helmet (100) includes a helmet (102) to be worn by the user for protection and is operatively connected to an ignition interlock unit (104). The ignition interlock unit (104) detects if the helmet is worn by the user and stops the vehicle from starting unless the helmet is worn. In an embodiment, the ignition interlock unit (104) is connected to a retention unit of the helmet (102). The retention unit includes a buckle and a guard to detect if the helmet is worn by the user. The retention unit also includes an extendable stripe that is used to adjust the length of the retention unit so that the helmet is fixed to the user's face. The retention unit is considered closed when the buckle and the guard are connected with each other. The retention unit is considered open when the buckle and the guard are not connected with each other. Furthermore, a microcontroller (110) sends a signal via a communication network (106) to the ignition interlock unit (104) to lock the ignition of the vehicle if the helmet is not worn by the user.

[0028] The smart helmet (100) includes a plurality of sensors (108) operatively connected to the helmet (102). The plurality of sensors (108) includes, but is not limited to, an accelerometer, a gyroscope, an ultrasonic sensor, and a location sensor. The location sensor is used to provide GPS location of the user. The plurality of sensors (108) is used to capture a sensor data, an operation data from the ignition interlock unit (104), and a location data. The plurality of sensors (108) is used to detect obstacles on the road or pathway. The helmet (102) is equipped with the plurality of sensors (108) designed to monitor the surrounding environment for potential hazards. The ultrasonic sensor detects proximity to obstacles and measure impact forces, providing real-time feedback to the user.

[0029] The smart helmet (100) is equipped with the microcontroller (110) integrated in the helmet (102). The microcontroller (110) is configured to receive the sensor data, the operation data from the ignition interlock unit (104), and the location data. The ultrasonic sensor plays a crucial role in measuring the distance to nearby objects, allowing the smart helmet (100) to assess potential hazards in real-time. The ultrasonic sensor is used to send ultrasonic waves from the helmet (102) that are reflected back if hit by an obstacle. The sensor data includes information related to the ultrasonic waves, which is analyzed by the microcontroller (110) to determine the proximity of obstacles and assess the risk of collision. This analysis enables the helmet (102) to provide real-time feedback to the user, enhancing situational awareness while riding.

[0030] The microcontroller (110) acts as a central processing unit, providing instructions and processing power to the smart helmet (100). The microcontroller (110) is configured for high-speed performance to efficiently complete tasks. In an embodiment, the microcontroller (110) retrieves instructions stored in memory in real time. The memory includes any computer-readable medium known in the art, such as volatile memory (e.g., RAM) and/or non-volatile memory (e.g., EPROM, flash memory, etc.). The memory is configured to store data and ensure data access by the smart helmet (100). The memory stores necessary data and configurations of the smart helmet (100). It is used to create and store databases, including but not limited to MySQL databases, hierarchical databases, NoSQL databases, cloud databases, object-oriented databases, centralized databases, and distributed databases. In an embodiment, the memory or a database is present alongside or inside the smart helmet (100).

[0031] The microcontroller (110) performs analysis of the sensor data, the operation data, and the operation data to identify approaching obstacles. In an embodiment, proximity to obstacles is detected through a combination of ultrasonic and infrared sensors, allowing the smart helmet (100) to calculate distances accurately and respond accordingly. In an embodiment of the present invention, the microcontroller (110) utilizes algorithms to process the data received from the plurality of sensors (108), allowing it to assess the environment in real-time.

[0032] The microcontroller calculates obstacle distance to alert the user if the distance is in the danger range. The distance to potential hazards is continuously monitored, enabling the smart helmet (100) to provide timely alerts to the user. This feature enhances safety by ensuring that the wearer is aware of their surroundings, particularly in high-risk environments. The distance data is processed by the microcontroller (110) to detect a danger range for the vehicle or the user. The danger range is defined as the threshold distance within which an obstacle poses a significant risk, prompting the smart helmet (100) to activate alert signal. If the distance is higher than the threshold distance, the smart helmet (100) does not take any action or generate any alerts. Furthermore, if the distance is less than the threshold distance, the smart helmet (100) activates an alert unit (112) to activate an alert signal.

[0033] The alert unit (112) provides a visual alert and audio alert on a communication device (118) when an obstacle is detected within a danger range. The visual alert is displayed on a user interface of the communication device (118), while the audio alert is played through a speaker on the communication device (118).

[0034] The user interface displays content to the user on the communication device (118). It serves as the point of interaction between the user and the smart helmet (100). The user interface includes, but is not limited to, graphical user interface, voice, touch, and command-line interfaces. For example, the user interacts with the user interface on the communication device (118). Furthermore, the user interface enables the smart helmet (100) to communicate with other computing devices, such as web servers and external data servers. The user interface is used by the user to view, interact with, and monitor the usage of the helmet on the communication device (118), which is transmitted via the communication network (106).

[0035] The communication device (118) is a portable or handheld device that the user employs to interact with the smart helmet (100). In one embodiment, the communication device (118) includes a mobile phone, tablet, desktop, and the like. The communication device (118) is used by the user to interact with, view approve, delete, upload, or modify information available related to the smart helmet (100). Further, the communication device (118) is used by the user to view final transaction, which is displayed on the user interface. In some embodiments, the communication device (118) includes additional components such as a camera, microphone, GPS, location sensor and speaker. The communication device (118) also includes the user interface that is used by the user to view the data related to helmet usage. The smart helmet (100) features a built-in communication network (106) that allows for seamless connectivity with other devices, enhancing situational awareness.

[0036] The communication network (106) enables the communication device (118) to connect with the helmet (102) and other components of the smart helmet (100). Additionally, the communication network (106) serves as a medium for transferring data or signals between a server and the smart helmet (100). This may include wired, wireless, infrared, radio frequency (RF), or similar technologies. The communication network (106) may function as either a dedicated network or a shared network. A shared network represents an association of various network types that utilize multiple protocols, such as Hypertext Transfer Protocol (HTTP), Transmission Control

Protocol/Internet Protocol (TCP/IP), and Wireless Application Protocol (WAP), to facilitate communication. Furthermore, the communication network (106) includes a variety of network devices, such as routers, switches, bridges, servers, and computing devices. It may also provide access to a storage device located on a client-side computer, a host-site server, cloud infrastructure, or a combination thereof. These storage options may include one or more local and remote storage media, such as memory storage devices, a database, and the like.

[0037] The communication network (106) is a wireless mobile network. In an embodiment of the present invention, it may be a wired network with finite bandwidth. In an embodiment, the communication network (106) combines both wireless and wired networks to optimize transmission throughput and latency. The communication network (106) can be implemented in various forms, such as an intranet, a local area network (LAN), a wide area network (WAN), the internet, or similar configurations. In another embodiment, the communication network (106) may be an optical fiber network offering high bandwidth, enabling rapid data transmission with minimal connection drops. Furthermore, the communication network (106) includes a set of channels, each with a finite bandwidth determined by the network's capacity. In one embodiment, the communication network (106) enables the smart helmet (100) to access the internet for transmitting and receiving data from the communication device (118) and other components.

[0038] Furthermore, the smart helmet (100) includes a rechargeable battery (116) to supply power to the microcontroller (110), the plurality of sensors (108), and the communication device (118). The rechargeable battery (116) is designed for extended use, allowing for multiple hours of operation on a single charge, ensuring that users can rely on the smart helmet (100) for prolonged activities without interruptions. In an embodiment, the rechargeable battery (116) is charged via a standard USB-C port, enabling convenient recharging options and compatibility with various charging devices, while also featuring an energy-efficient design that minimizes power consumption during idle periods.

[0039] In an embodiment, the smart helmet (100) integrates with GPS functionality, allowing for precise tracking of the user location. This can be crucial for emergency responders in case of an accident. Furthermore, the smart helmet (100) can log data related to the user riding patterns and incidents, contributing to a comprehensive safety profile that can be reviewed for improvements in riding behavior and helmet performance.

[0040] In an embodiment, the microcontroller (110) can integrate data from various sources, such as GPS and accelerometers, to offer comprehensive situational awareness. By analyzing this information, the smart helmet (100) can also adapt its notifications based on the user's speed and activity level, ensuring that alerts are relevant and actionable. Furthermore, the microcontroller (110) supports connectivity with external devices, allowing for seamless communication and data sharing, which can further enhance the functionality and user experience of the smart helmet (100). Additionally, the integration of GPS enables precise location tracking, ensuring that emergency responders can reach the user quickly and efficiently. This comprehensive approach not only enhances the safety of the rider, but also provides peace of mind to their loved ones, knowing that help can be summoned promptly in the event of an accident.

[0041] In an embodiment, the microcontroller (110) detects the impact on the vehicle and determines the severity of the situation. It then decides whether to activate additional safety features, such as automatic emergency calls or alerts to nearby vehicles. The smart helmet (100) automatically detects collisions and impact intensity. In an embodiment, the plurality of sensors (108) is used to detect the impact of a collision or accident on the helmet (102). The impact of the collision or accident is used to provide an alert to the emergency contact of the user based on the impact. If the impact of the collision is above a threshold level, the microcontroller (110) sends a signal to a messaging module (114). The messaging module (114) sends an SMS alert to the emergency contact of the user if the microcontroller (110) detects that the vehicle collision or accident impact is higher than the predefined threshold based on the sensor data. The messaging module (114) transmits alerts and location data to emergency contacts for immediate assistance. Additionally, the microcontroller (110) can activate visual or auditory alerts to warn the user of imminent dangers, ensuring a proactive approach to safety.

[0042] In an embodiment, the smart helmet (100) may include a server for managing, storing, and processing the data and information present in the smart helmet (100). Generally, the server is a computer program or device that provides functionality to other programs or devices. It oversees the operations and tasks performed by the smart helmet (100) and stores the instructions necessary for various functions. The server provides resources, data, storage capacity, and programs to facilitate these operations. It also supports functionalities such as data and resource sharing among multiple clients and performing computations on behalf of the clients. The server may provide additional resources, such as cloud-based processing or updates, enhancing the overall performance and adaptability of the smart helmet (100). Additionally, it would be apparent to those skilled in the art that the smart helmet (100) may connect to multiple servers to enhance scalability and performance.

[0043] **FIG. 2** illustrates a flowchart for obstacle detection by the smart helmet (100). The smart helmet (102) includes the ultrasonic sensor from the plurality of sensors (108) mounted on the front of the smart helmet (100). The ultrasonic sensor sends out ultrasonic waves in front of the vehicle which are reflected back when restricted by the obstacle (202). The reflected waves from the obstacle (202) are sensed by the ultrasonic sensor. The received ultrasonic wave information is shared with the microcontroller (110) as the sensor data. The microcontroller (110) analyses the sensor data to identify the distance of the obstacle (202) from the vehicle in real-time. The microcontroller (110) works to detect if the obstacle (202) is in the danger range or safe range. The safe range is considered when the distance of the obstacle (202) is higher than the pre-defined threshold. If the obstacle (202) is in the safe range, the microcontroller (110) does not take any action and continues to monitor the road or pathway. If the obstacle (202) is in the danger range, the microcontroller (110) activates the alert unit (112) to generate the visual alert and audio alerts about the approaching obstacle on the communication device (118).

[0044] From the above description, it will be appreciated that many variations are possible in the above invention. Although every possible embodiment has been disclosed in its preferred form(s), the specific embodiment as disclosed and illustrated herein is not to be considered in a limiting sense, because numerous variations are possible. The subject matter of the inventions includes all novel and non-obvious combinations and sub-combinations of the various elements, features, functions, and/or properties disclosed herein. The following claims particularly point out certain combinations and sub-combinations regarded as novel and non-obvious. Inventions embodied in other combinations and sub-combinations of features, functions, elements, and/or properties may be claimed in applications claiming priority from this or a related application. Such claims, whether directed to a different invention or to the same invention, and whether broader, narrower, equal, or different in scope to the original claims, are also regarded as included within the subject matter of the inventions of the present invention.

We claim:

1. A smart helmet (100) comprising:

   a helmet (102) to be worn by a user for protection and operatively connected to an ignition interlock unit (104) to detect if the helmet (102) is worn by the user;

   a plurality of sensors (108) operatively connected to the helmet (102) to detect obstacles and impacts on the helmet (102);

   a microcontroller (110) integrated in the helmet (102) and configured to receive a sensor data, operation data from the ignition interlock unit (104), and location data, and perform analysis to identify approaching obstacles; and

   an alert unit (112) to provide a visual alert and an audio alert on a communication device (118) when an obstacle is detected in a danger range.

   wherein the microcontroller (100) sends a signal via a communication network (106) to the ignition interlock unit (104) to lock the ignition of a vehicle if the helmet (102) is not worn by the user.
2. The smart helmet as claimed in claim 1, wherein the plurality of sensors (108) comprises an accelerometer, a gyroscope, an ultrasonic sensor, and location sensor.
3. The smart helmet as claimed in claim 1, further comprising a rechargeable battery (116) to supply power to the microcontroller (110), the plurality of sensors (108), and the communication device (118).
4. The smart helmet as claimed in claim 1, further comprising a retention unit with a buckle and a guard to detect if the helmet (102) is worn by the user, wherein the retention unit is considered as closed when the buckle and the guard are connected with each other.
5. The smart helmet as claimed in claim 1, further comprising a messaging module (114) to send an SMS alert to emergency contact of the user if the microcontroller (110) detects vehicle collision based on the sensor data.
6. The smart helmet as claimed in claim 1, wherein the microcontroller (110) analyses the ultrasonic waves from the ultrasonic sensors of the plurality of sensors (108) and calculate obstacle distance to alert the user if the obstacle distance is in the danger range.

7. The smart helmet as claimed in claim 1, wherein the visual alert is shared on a user interface of the communication device (118) and the audio alert is shared via a speaker on the communication device (118).
8. The smart helmet as claimed in claim 1, wherein the microcontroller (110) monitors helmet usage of the user and display it on a user interface of the communication device (118).

ABSTRACT

A SMART HELMET

A smart helmet (100) which includes a helmet (102) to be worn by a user for protection and operatively connected to an ignition interlock unit (104) to detect if the helmet is worn by the user. A plurality of sensors (108) operatively connected to the helmet (102) to detect obstacles and impacts on the helmet (102). A microcontroller (110) integrated in the helmet (102) and configured to receive a sensor data, operation data from the ignition interlock unit (104), and location data, and perform analysis to identify approaching obstacles. An alert unit (112) to provide a visual alert and an audio alert on a communication device when an obstacle is detected in a danger range. The microcontroller sends a signal via a communication network (106) to the ignition interlock unit (104) to lock the ignition of a vehicle if the helmet is not worn by the user.